NLP Language Modelling exercises

Miryam de Lhoneux

Q1 Consider the following corpus of 3 sentences:

- <s> I am Sam </s>
- <s> Sam I am </s>
- <s> I saw a unicorn </s>

Q1.1 Compute all bigram probabilies by filling table 1 where the rows correspond to the first word of the bigram (w_{n-1}) and the columns correspond to the second word (w_n).

	I	am	Sam	saw	a	unicorn	</s>
<s>							
I							
am							
Sam							
saw							
a							
unicorn							

Table 1: Bigram probabilities

🔑 **Q1.1** See table 2.

	I	am	Sam	saw	a	unicorn	</s>
<s>	$\frac{2}{3} = 0.67$	0	$\frac{1}{3} = 0.33$	0	0	0	0
I	0	$\frac{2}{3} = 0.67$	0	$\frac{1}{3} = 0.33$	0	0	0
am	0	0	$\frac{1}{2} = 0.5$	0	0	0	$\frac{1}{2} = 0.5$
Sam	$\frac{1}{2} = 0.5$	0	0	0	0	0	$\frac{1}{2} = 0.5$
saw	0	0	0	0	1	0	
a	0	0	0	0	0	1	
unicorn	0	0	0	0	0	0	1

Table 2: Bigram probabilities

Q1.2 Based on these probabilities, compute the probability of the following sentences:

- <s> I am </s>
- <s> I am a unicorn </s>
- <s> I saw a whale </s>

🔑 **Q1.2**

- <s> I am </s> : $\frac{2}{3} \times \frac{2}{3} \times \frac{1}{2} = 0.22$
- <s> I am a unicorn </s> : $\frac{2}{3} \times \frac{2}{3} \times 0 \times 1 \times 1 = 0$
- <s> I saw a whale </s> : undefined

Q1.3 Generate a sentence using our bigram model, selecting the most probable bigram at each step (you choose how to break ties).

🔑 **Q1.3**

- <s> I am </s>
- <s> I am Sam </s>
- <s> I am Sam I am </s>
- <s> I am Sam I am Sam I am Sam ... *(possible infinite loop!)*

Q1.4 ✏️ Smoothe your bigram probabilities with add-one smoothing from the earlier exercise. Recompute the probabilities for the sentences of **Q1.2**.

🤔 How are the probabilities affected by smoothing? Would add-k with a value of k between 0 and 1 be better? Why? (It can be informative to try this exercise with other values of k to form a clearer intuition).

🔑 **Q1.4** The bigram probabilities are in table 3.

	I	am	Sam	saw	a	Unicorn	</s>
<s>	$\frac{3}{10} = 0.3$	$\frac{1}{10} = 0.1$	$\frac{2}{10} = 0.2$	$\frac{1}{10} = 0.1$	$\frac{1}{10} = 0.1$	$\frac{1}{10} = 0.1$	$\frac{1}{10} = 0.1$
I	$\frac{1}{10} = 0.1$	$\frac{3}{10} = 0.3$	$\frac{1}{10} = 0.1$	$\frac{2}{10} = 0.2$	$\frac{1}{10} = 0.1$	$\frac{1}{10} = 0.1$	$\frac{1}{10} = 0.1$
am	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{2}{9} = 0.22$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{2}{9} = 0.22$
Sam	$\frac{2}{9} = 0.22$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{1}{9} = 0.11$	$\frac{2}{9} = 0.22$
saw	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{2}{8} = 0.25$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$
a	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{2}{8} = 0.25$	$\frac{1}{8} = 0.12$
unicorn	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{1}{8} = 0.12$	$\frac{2}{8} = 0.25$

Table 3: Bigram probabilities smoothed with add-one.

Here are the probabilities of the sentences:

- <s> I am </s> : $\frac{3}{10} \times \frac{3}{10} \times \frac{2}{9} = 0.02$
- <s> I am a unicorn </s> : $\frac{3}{10} \times \frac{3}{10} \times \frac{1}{9} \times \frac{2}{8} \times \frac{2}{8} = 0.000625$
- <s> I saw a whale </s> : still undefined

Q1.5 ✏️ Calculate the interpolated probabilites of the sentences of **Q1.2** with the bigram model of **Q1.1**, a MLE unigram model, and $\lambda_1 = \lambda_2 = 0.5$.

🤔 How do the results compare to add-one smoothing? How would altering the λ weights alter the results? (It can be informative to try this exercise with other λ weights).

I	am	Sam	saw	a	Unicorn	</s>
$\frac{3}{13} = 0.21$	$\frac{2}{13} = 0.13$	$\frac{2}{13} = 0.13$	$\frac{1}{13} = 0.07$	$\frac{1}{13} = 0.07$	$\frac{1}{13} = 0.07$	$\frac{3}{13} = 0.21$

Table 4: MLE unigram probabilities

🔑 **Q1.5** First, we need to calculate the unigram probabilities, see table 4.
Now we can calculate the interpolated probabilities:

- <s> I am </s> :

$$(\frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{3}{13}) \times (\frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{2}{13}) \times (\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{3}{13}) = 0.0673$$

- <s> I am a unicorn </s> :

$$(\frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{3}{13}) \times (\frac{1}{2} \times \frac{2}{3} + \frac{1}{2} \times \frac{2}{13}) \times (\frac{1}{2} \times 0 + \frac{1}{2} \times \frac{1}{13}) \times (\frac{1}{2} \times 1 + \frac{1}{2} \times \frac{1}{13}) \times (\frac{1}{2} \times 1 + \frac{1}{2} \times \frac{3}{13}) = 0.002$$

- <s> I saw a whale </s> : still undefined